

The service is designed to be intuitive in manner and conduct. The user can use regular expressions (search for patterns of words rather than keywords) to search for more precise results. Searches can be restricted by language, license, or filename, and the user can trace the link right back to its originating package and Web page.

Google has also released the API for Code Search, which utilizes Google's GDATA API format (<http://code.google.com/apis/gdata/index.html>).

One important thing to note is that Google Code Search is different from Google Search (the standard search engine), and therefore, the results of Code Search results are not integrated into the main Google index.

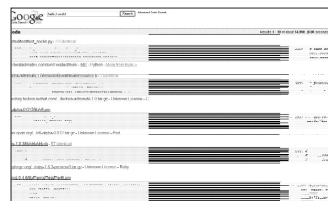
2.2 How To Search

The Code Search interface is similar to the standard Google search page. However, the first thing anyone would notice is the set of syntax instructions and examples given right under the search function.

1. **regex**—a regular expression is meant to search for a pattern of words rather than a set of keywords. For example, “hello,\world”. When this is used, the search engine opens up the result page that displays code corresponding to that requirement.



The Google Code Search Syntax



A Google Code Search Result

2. **Exact string**—to search for an exact string, enclose the search string within double quotes. For example, “hello world”
3. **file:regex**—this searches only in files or directories matching the regular expression.
4. **package:regex**—this searches for packages containing the specified expression. This presents a list of archives in the ZIP or TAR formats, for example.
5. **lang:regex**—this crawls through the repositories for code written in the specified language. For example, you could use the search string “lang:c++ hello world”.
6. **license:regex**—this searches for code available under a particular license. To cite an example, the search string “license:bsd hello world” gets you all code licensed under the BSD license, containing the words “hello” and/or “world”.

2.3 Advanced Code Search

The Advanced Code Search page is for more specific search queries. For instance, it provides a list of the programming languages that correspond to your search requirement. Your results will turn up even if the language you’re looking for is not listed—just select “any language.”

2.4 The Benefits of Code Search

Code Search allows coders to refer to a number of sources, compare their own code or to get ideas on how to tackle a particular issue. According to programmers, this removes roadblocks that existed earlier, since it provides different ways of approaching a particular programming problem. Code Search could also allow code auditors to warn people much sooner of vulnerabilities in a program. A cen-